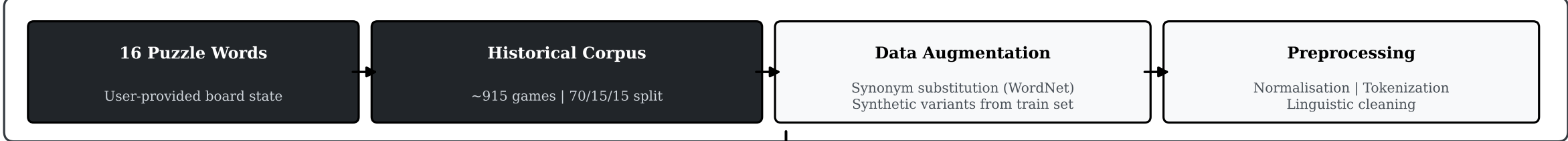


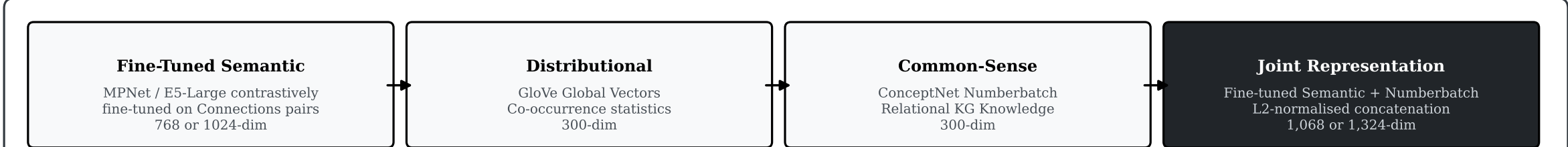
Proposed Architecture: NYT Connections NLP Solver with Multi-Signal Fusion

Contrastive Semantic Adaptation, Phonetic/Morphological Signals, and Beam Search Decoding

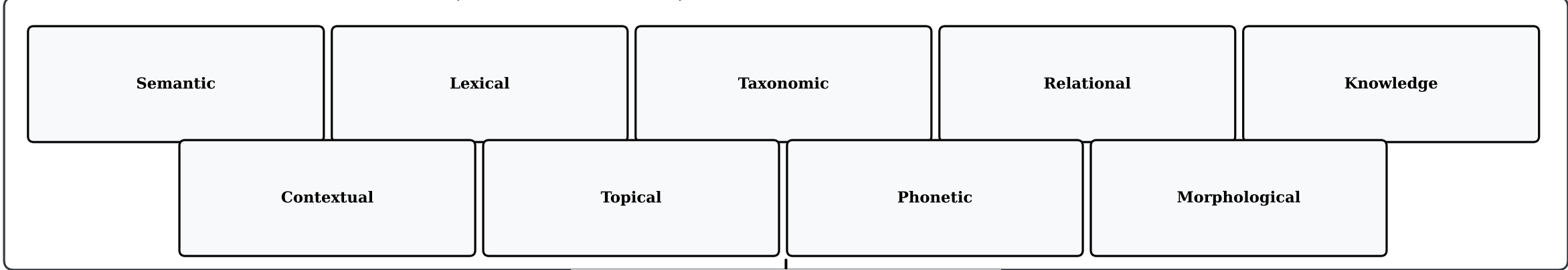
STAGE I: INPUT & DATA PREPARATION



STAGE II: WORD REPRESENTATION



STAGE III: SIMILARITY COMPUTATION (Nine 16x16 Matrices)



Double-Centering Normalisation: $\tilde{S}_{ij} = S_{ij} - \tilde{S}_{i.} - \tilde{S}_{.j} + \tilde{S}_{..}$

STAGE IV: DOWNSTREAM PIPELINES (Inference & Training Paths)



Legend:

Input / Output Node

Processing Logic

Pipeline Boundary (Dashed = Training Stage)

Figure 1: Research Architecture for the NYT Connections NLP Solver. The system integrates (i) contrastive fine-tuning of semantic embeddings, (ii) a multi-signal similarity matrix architecture including phonetic and morphological patterns, (iii) an extended feature set (~60 dimensions) for supervised ensemble classification, and (iv) look-ahead beam search decoding with uncertainty filtering to mitigate early commitment errors.